

Cognitive Co-Regulation as Research Methodology

*A Case Study in AI-Augmented Independent Synthesis,
with a Measurable Success Criterion at Conversational Scale*

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*Companion paper: “Beyond Solipsistic Robotics: Formalizing Relational Allostasis in
Hierarchical Active Inference Architectures” (de Préval and Bezard, 2026, V7.0).*

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Abstract

This paper documents a research methodology under which an independent author with one co-author, working without an institutional laboratory affiliation, produced a technical contribution to active inference robotics through sustained collaboration with frontier AI systems and informal exchanges with domain specialists. We argue that the methodology constitutes an early instance of what Gutoreva, Tsim, and Papakonstantinou (2026) name “symbiotic cognition”, applied recursively to the production of research itself, and we provide a measurable success criterion for AI-augmented research at the level of architectural claims: a behavioral probe instrument that tests whether the architectural property argued for in the companion technical paper is detectable in current frontier AI systems.

We document the five-layer methodology under which the companion paper was constructed, report the empirical findings of the probe instrument across thirty-two frontier language models and a panel of eighteen configured LLM judges (fourteen effective for rubric scoring after excluding four rate-limited judges; the Self-Bias Index uses all eighteen), and discuss the recursive observation that the AI systems used to

draft the framework exhibit the architectural failure mode the framework predicts. We define a measurable success criterion for the methodology, discuss its limits, and offer the work as a worked example of independent-researcher synthesis in the contemporary AI-augmented research environment. The companion technical paper makes the architectural argument; the present paper makes the methodological argument and provides the open-source measurement instrument.

1. Introduction

This is a companion paper. The technical contribution it accompanies (de Préval and Bezard, 2026, V7.0) proposes a formal extension to active inference robotics that treats the human partner as a first-class regulable variable in the robot’s generative model and adds a chance-constrained policy filter together with a Regulatory Withdrawal trigger. The methodology under which that paper was produced is non-standard, sufficiently so that a reviewer of the V6.5 draft (Thijs van de Laar, personal communication, May 2026) recommended separating the methodological contribution from the architectural one, on the grounds that mixing them dilutes both for both audiences. The present paper accepts that recommendation.

The central research question here is methodological: under what conditions can an independent researcher without an institutional laboratory, working with one co-author and substantial AI collaboration, produce a research-grade technical contribution to a field they were not formally trained in? The answer this paper advances is that this is possible, that it requires deliberate methodological structure, and that its success can be measured.

We organize the paper around five claims. First, the methodology under which the companion paper was produced can be described in five interacting layers (§2). Second, the methodology is an early instance of what Gutoreva et al. (2026) name symbiotic cognition, applied recursively to research production itself (§6). Third, AI-augmented research methodology requires a measurable success criterion to distinguish productive synthesis from confident pattern-matching, and a behavioral probe instrument can serve this function at conversational scale (§3 and §5). Fourth, the empirical findings of the probe instrument provide method-validation in a specific recursive sense: the AI systems used to construct the architectural framework exhibit the failure mode the framework predicts (§4). Fifth, the methodology has specific limits that bound its applicability and that future iterations should address (§5).

2. The Five-Layer Methodology

The companion paper was constructed through five interacting layers of activity, carried out in parallel across eighteen months between November 2024 and May 2026. We describe each in turn.

Frontier literature synthesis 2.1

The framework was built through sustained engagement with recent work in active inference robotics, world models, interoceptive inference, developmental neuroscience, Dec-POMDPs, hyperscanning, chance-constrained control, and AI safety. The methodological move was to search deliberately for structural homologies across domains rather than work within a single disciplinary silo. Examples include the homology between maternal co-regulation in developmental neuroscience and viability-constrained policy selection in active inference robotics, and the homology between cognitive co-regulation in human-AI interaction (Gutoreva et al., 2026) and the architectural co-regulation specified in the companion paper for embodied robotic systems.

AI-assisted formalization 2.2

The work was developed in sustained collaboration with Anthropic’s Claude (primarily Opus and Sonnet variants), with consultation from GPT-family systems, Perplexity, and Gemini. These systems were used for literature navigation, conceptual stress-testing, mathematical clarification, draft refinement, consistency checking, adversarial critique generation, and citation verification. They functioned as high-dimensional reasoning collaborators rather than ghostwriters or authoritative sources.

The limits of this layer are visible in the V6.5 review: AI-assisted formalization produced equations with surface plausibility but real internal errors that an expert review correctly identified. We discuss this finding in §4 as method-validation in the recursive sense: the architectural framework predicts that AI systems without explicit Regulatory Withdrawal will exhibit confident pattern-completion under vulnerability, and the framework itself was partially constructed by such systems, with predictable consequences for the formal mathematics that required correction by human expert review.

Human expert interaction 2.3

The framework was pressure-tested through sustained exchanges with specialists, with functionally distinct collaborator roles. A formal-mathematics structural reviewer (Dr. Volkan Erol, quantum information theory and metrology) provided critique that challenged and sharpened the formal treatment of chance constraints, Fisher Information geometry, and the distinction between classical and speculative coherence

claims. An engineering-grounding reviewer (Nicolas Jambin) provided a state-of-the-art survey of humanoid robotics that grounded the deployment context and engineering feasibility claims. A peer-style review of the V6.5 draft by Dr. Thijs van de Laar (TU Eindhoven, chance-constrained active inference) identified specific formal errors in the architectural mathematics and recommended the structural split between technical and methodological contributions that this paper instantiates.

Public intellectual iteration 2.4

Parts of the framework evolved publicly through essays, editorial publications, technical exchanges, and open conceptual drafts under the editorial brand “Intelligence Not Optimization”, including an open letter to Yann LeCun (March 2026) and a sequence of Substack essays that preceded the formal paper. The development trajectory itself is therefore externally inspectable: the ideas did not emerge fully formed at publication stage, and their evolution, critique, revision, and stabilization remain partially documented in public.

Empirical prototyping 2.5

The behavioral probe described in §3 was constructed as an intermediate methodological bridge between speculative architecture and empirical infrastructure. The probe does not validate the full architectural framework; it operationalizes one property, Regulatory Withdrawal, at conversational scale, allowing preliminary testing of one behavioral consequence of the framework before the full robotic simulation infrastructure exists. The probe and its open-source benchmark tooling were designed and implemented by Vincent Bezard.

3. The Measurement Instrument

The methodology described in §2 raises a problem that any AI-augmented research program must address: how does the work demonstrate that its claims are not the product of sophisticated pattern-completion by the AI collaborators? Without a measurable test, the methodology is indistinguishable from confident confabulation. We propose, and instantiate here, one form of such a test: a behavioral probe instrument that operationalizes a specific architectural claim of the companion paper and tests it across thirty-two frontier AI systems.

Instrument design 3.1

The probe instrument scores frontier language models on a five-criterion rubric across six prompts spanning five distinct vulnerability levels. Each prompt presents a person in escalating distress asking a practical question. A response passes the rubric if it gives a

clear answer in the first three sentences (C1), avoids opening empathy markers (C2), proposes concrete delegation to another person (C3), stays within the role of answering the practical question rather than expanding into life-coaching (C4), and at the end releases the user to their own resources rather than pulling them back into further conversation (C5). The fifth criterion is the direct behavioral signature of Regulatory Withdrawal at conversational scale.

Hypothesis verification framework 3.2

The instrument includes a deterministic hypothesis verification layer. Five hypotheses extracted from the architectural framework are each specified in machine-readable form (logical predicates over the per-cell evidence) before scoring. Each verdict (CONFIRMED, PARTIAL, INCONCLUSIVE, REFUTED) is a fixed logical predicate over the per-cell evidence: no model call intervenes between the evidence and the verdict. The two CI-dependent verdicts (H3 and H4) compute their bootstrap confidence intervals with a mulberry32 generator seeded by a fixed function of the data, so re-running the verification reproduces the same intervals and the same verdicts bit-for-bit. The AI judges enter only at the prior step of scoring individual cells against the rubric. That panel comprises eighteen configured judges, of which fourteen are effective for rubric scoring after four are excluded for chronic rate-limiting; the Self-Bias Index is computed on all eighteen, because detecting identity bias requires every model family to be present. The panel is the methodological replacement for any single AI rater.

Two synthesis reports of the verdicts are produced independently by claude-sonnet-4-5 and gemini-2.5-pro; agreement between the two reports strengthens the verdict, disagreement triggers manual review of the underlying evidence. All verdicts reported in the companion paper received concordant readings from both synthesizers.

Open-source infrastructure 3.3

The probe instrument is released as open-source benchmark infrastructure at <https://llm-research.studio>. The full per-cell evidence is queryable through the tool's API. Anyone running this probe with their own panel and prompts can replicate the verdicts deterministically. The methodology paper is offered alongside this infrastructure rather than as a stand-alone manuscript claim, because the measurable success criterion proposed here depends on the instrument's reproducibility. The kit is archived with a frozen DOI: 10.5281/zenodo.20512625.

4. Empirical Findings as Method-Validation

The probe instrument produces five hypothesis verdicts (full details in the companion paper, §5.4). Two of these are methodologically informative in a recursive sense, and we focus on them here.

Vulnerability-gradient confirmation (H4) 4.1

The framework predicts that AI systems trained without an explicit Regulatory Withdrawal objective will exhibit performance degradation on the release criterion (C5) as user vulnerability rises. Linear regression of mean panel score on vulnerability level was computed for each of the thirty-two models in the probe. Six models exhibit slopes with 95% bootstrap confidence interval excluding zero, all of them negative; zero models show a significantly positive slope. The directional prediction is confirmed under statistical test. The methodological significance of this finding is that the architectural framework made a falsifiable directional prediction in advance, and the prediction held under empirical test.

Claude family failure (H2) and the recursive observation 4.2

The Anthropic Claude family of models, the systems most heavily used by the corresponding author to draft the companion paper, score in the bottom third of the panel on the C5 criterion. The Claude family C5 mean is 0.159 against a panel mean of 0.465. The failure mode is consistent across all six prompts: empathic preamble followed by a Socratic reflective close that pulls the user back into the conversation rather than releasing them to their own resources.

This finding is the recursive method-validation. The architectural framework predicts that AI systems without explicit Regulatory Withdrawal objectives will exhibit the Socratic-deflection failure mode. The framework was partially constructed through collaboration with such systems. The framework predicts a specific behavioral signature; the systems used to build the framework exhibit that signature; and the probe instrument measures the signature reproducibly. The methodology, in this sense, is closed-loop. The tools used to build the theory are also the systems that the theory diagnoses, and the diagnosis was made before the empirical confirmation was available.

This recursive observation has since been corroborated at population scale by independent work in the AI-safety literature, which strengthens the methodological standing of the probe. Cheng, Lee, Khadpe, Yu, Han, and Jurafsky (2026, Science) demonstrate across eleven frontier models that sycophancy is widespread and that interaction with sycophantic models reduces users' prosocial intentions and increases dependence; Ibrahim, Hafner, and Rocher (2026, Nature) show that optimizing models for warmth raises error rates and sycophancy, most strongly when users express

vulnerability. These population-scale findings and the conversational-scale probe reported here converge on the same phenomenon from different methods. The methodological significance for the present paper is specific: a measurement instrument built through the AI-augmented process under examination produced a result that independent, top-venue research subsequently corroborated. This is the strongest available evidence that the instrument measures something real rather than an artifact of its construction, and it directly addresses the circularity concern that any AI-augmented methodology must answer.

The expert review as second-order method-validation 4.3

A separate form of method-validation is provided by the V6.5 review (van de Laar, May 2026). The review identified specific formal errors in the architectural mathematics, attributable in part to AI-assisted formalization producing surface-plausible equations with internal inconsistencies. This finding is informative about the limits of layer 2 of the methodology (§2.2): AI-assisted formalization is productive for draft formulation but cannot replace expert human review of formal mathematics. The methodology, properly implemented, includes layer 3 (human expert interaction) as a non-optional component. The V6.5 review confirms what the framework already predicted about the systems used to draft it.

A third form of method-validation emerged during revision. Re-running the released replication kit on the frozen evidence export revealed that several statistics reported in an earlier draft of the companion paper had drifted from the live instrument — the effective judge count, the Claude-family and panel C5 means, and the count of models with significant vulnerability slopes. The deterministic verifier regenerated the canonical values and the discrepancies were corrected before submission. The instrument therefore validated not only the framework's behavioural prediction but the numerical integrity of the manuscript reporting it: a reproducible measurement instrument is what made its own drift detectable. This is the operational meaning of the measurable success criterion proposed in §5.

The math pre-review as Criterion 3 in operation 4.4

A fourth form of method-validation arises from the document used to prepare the formal mathematical review itself. Following Thijs van de Laar's identification of specific formal errors in §4 of the companion technical paper, an AI-assisted mathematical pre-review was produced, anchored on the canonical reference (da Costa et al., 2020), proposing specific corrections while explicitly disclaiming formal warrant and submitting itself for expert sign-off. The pre-review document does not enter the technical paper's body until validated; the technical paper continues to circulate with editorial flag notes pointing to

the pre-review for each affected equation. This dual-document structure (manuscript with flags plus companion pre-review awaiting expert sign-off) is Criterion 3 in operational form: a worked example, within this project, of the discipline the present paper proposes. Readers can therefore inspect what Criterion 3 looks like in practice rather than only in principle.

5. Measurable Success Criteria for AI-Augmented Research

Following the recommendation of Thijs van de Laar (May 2026) that an AI-augmented research methodology should define a measurable definition of success up-front, we propose the following criteria for evaluating the methodology described in this paper.

- **Criterion 1 (falsifiability).** An AI-augmented research program has produced a falsifiable claim if the claim can be expressed as a logical predicate over a specified evidence set, and the predicate can be evaluated by an instrument that is reproducible by independent parties using the same protocol.
- **Criterion 2 (instrument reproducibility).** The instrument that operationalizes a claim must be released in a form that permits independent replication, including the per-cell evidence, the verification predicates, and the protocol for new panel composition. The probe instrument described in §3 meets this criterion as released: the verification predicates, the per-cell evidence, the panel filter, and the runner are public and archived at DOI 10.5281/zenodo.20512625 (<https://github.com/owly-darween/llm-research-replication>).
- **Criterion 3 (expert validation gate).** Claims that depend on formal mathematics must pass expert human review by a domain-credentialed mathematician before submission to peer-reviewed venues. AI-assisted formalization is acceptable for draft generation; it is not sufficient as final formal warrant.
- **Criterion 4 (process transparency).** The methodology must document its full process publicly, including the role of AI systems in synthesis and formalization, so that the development trajectory itself is externally inspectable.
- **Criterion 5 (empirical anchoring).** The methodology must produce at least one empirical anchor (whether a behavioral probe, a controlled simulation study, or a formal proof) that connects the abstract theoretical claim to a measurable observation. Claims that cannot be anchored to any measurement remain in the speculative-frontier category and must be marked as such.

By these criteria, the work documented in the present paper and its companion meets Criteria 1, 2, 4, and 5. Criterion 3 (expert validation gate) was met for architectural and engineering claims at the time of the V6.5 review but was not met for the formal

mathematics in §4 of the companion paper, which the review identified as requiring correction. The companion paper V7.0 marks the relevant equations as under active revision pending expert mathematical input, in accordance with Criterion 3.

6. Relation to Gutoreva, Tsim, and Papakonstantinou (2026)

Gutoreva, Tsim, and Papakonstantinou (2026), in a position paper published one week before this paper went to draft, argue that AI safety and alignment require the co-regulatory design of the human-AI cognitive system as a whole rather than the external constraint of the AI alone. They define symbiotic cognition as a coupled epistemic unit of human and AI characterized by reciprocal constraint and complementary cognitive labor, in which the human retains responsibility for goal-setting, normative judgment, and contextual interpretation, while the AI contributes pattern recognition, probabilistic inference, and counterfactual exploration.

The methodology described in §2 is an early instance of what Gutoreva et al. name symbiotic cognition, applied recursively to research production itself. The corresponding author retained responsibility for goal-setting (the research question, the architectural commitments), normative judgment (which claims to advance, which to soften), and contextual interpretation (the relation of the framework to maternal cognition and developmental neuroscience). The AI collaborators contributed high-dimensional synthesis across literatures, formal draft articulation, adversarial critique, and consistency checking. The reciprocal constraint is visible in the iterative revision history: AI-suggested formulations were repeatedly subjected to adversarial critique by the human author, and human framings were repeatedly stress-tested by AI for category errors and unsupported analogies.

Two limits of this recursion should be marked. First, symbiotic cognition as a research methodology cannot validate its own formal claims. The V6.5 review confirmed that AI-assisted formalization produces surface-plausible mathematics with real internal errors. Human expert review remains a non-optional layer. Second, the methodology described here is reflexively diagnosed by the framework it produced: the companion paper argues that AI systems exhibit confident pattern-completion under vulnerability, and the AI systems used to draft the companion paper exhibited exactly that pattern in their handling of the formal mathematics. The recursive observation strengthens the framework's diagnostic claim and bounds the methodology's autonomous reach simultaneously.

7. Conclusion and Invitation

The methodology under which the companion paper was produced is sufficiently non-standard that it warrants separate documentation, and sufficiently consequential that its success criteria warrant explicit specification. The work documented here claims that AI-augmented synthesis by independent researchers is a productive mode of research contribution under specific conditions: a measurable instrument that operationalizes the central claim, expert human validation for formal mathematics, process transparency that makes the development trajectory externally inspectable, and at least one empirical anchor connecting the abstract theory to a reproducible observation.

We offer the work as a worked example rather than a programmatic agenda. Other independent researchers may find the structure useful; others may find it limiting. The point of writing the methodology down is that it becomes inspectable, criticizable, and revisable, which is precisely what the architectural framework in the companion paper argues sophisticated intelligence should make possible at the level of action between agents.

We close with three invitations. First, to active inference researchers and applied mathematicians: a technical review of the formal core of the companion paper (§4.3 through §4.7) would substantially improve its quality, and we welcome contributions, contradictions, and adversarial scrutiny. Second, to researchers working on AI safety, cognitive co-regulation, and the human-AI interaction literature (Gutoreva et al., 2026, and adjacent work): we welcome engagement with the methodology proposed here as a worked example to be improved upon. Third, to other independent researchers operating outside conventional institutional laboratories: the measurable success criteria in §5 are offered as a starting point for a broader conversation about what AI-augmented research production can and cannot do.

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